

Advertising Cost of Living Improvements Through Childcare Costs

Term Project Final Paper

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DSC 640: Data Presentation and Visualization

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5/1/2025

Summary of Analysis:

My analysis began by noticing the inequity of both childcare costs and income across the U.S. This inspired the project of comparing family incomes and the relative cost of childcare to identify locations where different family's income may go further allowing them to live a better quality of life.

My first steps involved creating U.S. state heatmaps of median family income, median cost of center based childcare and relative childcare cost index (childcare cost proportion of income). From these heatmaps, I found Rhode Island to have a very high relative childcare cost while Kansas had the lowest index in the nation. This phenomenon was then found consistent across all age demographics of childcare. Meaning that if a family chose to move from Rhode Island to Kansas, they could save over \$42,000 over the course of the 5 years a child would require daycare.

Following this I set out to describe how location may affect relative childcare costs for single mothers. Creating a cost index finding the proportion of center-based childcare across all ages to median female income in each county. Williamsburg, Virginia had the highest relative childcare cost for single mothers while West Odessa, Texas has one of the lowest relative costs. Meaning that if a single mother moved between these two markets their disposable income could increase by nearly \$64,000 annually on median.

Finally, the cost of center-based childcare as a portion of median family income was found for all 50 states. Upon results the least expensive states were Kansas, South Dakota and Georgia. The most expensive states were Alaska, Rhode Island and Washington DC. These metrics were then compared across all three stages of childcare (infant, toddler and preschool)

and compared to the national averages. If a family moved from Alaska to Kansas, they could save 8.6 % of their annual income.

Story to Tell:

The story told in this project is that families may experience a better quality of life and less financial strain by moving their families from areas of high childcare cost relative to income to lower cost areas. This project also seeks to provide recommendations for some of the most financially distressed Americans such as single mothers.

This message is being delivered to these families courtesy of the low-cost childcare states and cities who seek to increase their states population and tax base. This will be a win-win scenario for these states and the families moving as the move promises a better quality of life through increased disposable income and the states and cities benefit from the increased tax base and population growth.

Audience:

For this project two distinct audiences have been identified. First being families that live in high childcare cost states and are currently paying for childcare or planning to soon. Due to the broadness of this audience, it is essential that the visualizations either remain clean and simple or are very well explained to deliver the message that the family's quality of life can improve by moving to a lower cost state.

The second target audience for this project is single mothers living in a high childcare cost city or county. This audience is very diverse in terms of data literacy; however, it can be assumed that all single mothers are extremely busy so any information conveyed must be done quickly and efficiently.

Mediums to Use:

Three mediums were chosen to deliver this message, the first of which being a Tik Tok or Instagram Reel. This being the largest reaching medium that is appealing to the largest audience, families living in high childcare cost areas.

The second medium is a listicle that is appealing to a similar audience. However, this is a more active medium and therefore the visualizations can be more descriptive and provide more information.

The final medium is an Instagram ad targeted at single mothers living in high childcare cost areas. This medium was chosen for single mothers because it is the fastest to the point and easy to consume for busy single mothers.

Design:

Color was a major consideration in visualization design for this project. Color in general was used sparingly across all visualizations to not distract the viewer from the data. In many circumstances only the colors green or red were used to convey high or low-cost values. National averages utilized grey to not distract from the data it was being compared to.

Gestalts principle of proximity was also used in grouped bar graph to help the viewer understand which data points belonged to which groups. Additionally, principles of closure and continuity were used to help the viewer differentiate between different arguments in the data or by removing borders to help them understand the relationship between visualizations.

Ethical Considerations:

Some ethical considerations for this project are that many of the arguments made are based on the median costs and incomes in each area. There is a potential that a family may choose to move and not experience the state or city's median costs or income. Potentially leaving them in a worse situation than they were before they moved.

Additionally, some of the visualizations display data points from a specific county. As visualizations are made on a smaller number of observations the risk of misrepresenting reality increases. This could also lead to single mothers moving and not experiencing the lifestyle gains portrayed in the Instagram ad.

Lessons Learned:

The most enjoyable part of this project was creating visualizations that may help families live more comfortable lives. I have seen firsthand how expensive some U.S. markets can be and visualizing that there are other options for families was very rewarding.

If I were to perform this project again, I may dig further into the variation of childcare costs and incomes in each area to describe the situation more fully. Leaning heavily on the median is a great start but more powerful and wholistic recommendations can be given if the spread of costs and incomes were considered.

References

National Database of Childcare prices. DOL. (n.d.).

<https://www.dol.gov/agencies/wb/topics/featured-childcare>